

# Number Power

Every graded set, every week — 4 weeks, 20 two-page sheets, with an answer key after each week.

WEEK 1

5 sets

WEEK 2

5 sets

WEEK 3

5 sets

WEEK 4

5 sets

## HOW TO USE IT

One page per day. Warm-Up is recall, Core is the graded tier, Challenge is never graded — the score box counts the graded items only. The same items appear in the app's Practice Bay, so a day can be done on paper or on screen, not both.

# How Big Is a Million

Read the place, then say what the digit is worth. Type digits only — no commas needed.

## Warm-Up 11 ITEMS • RECALL

1 How many tens in 100?

\_\_\_\_\_

2 How many hundreds in 1,000?

\_\_\_\_\_

3  $10 \times 100$

\_\_\_\_\_

4  $100 \times 100$

\_\_\_\_\_

5  $1,000 \times 10$

\_\_\_\_\_

6 How many tens in 50

\_\_\_\_\_

7 How many hundreds in 700

\_\_\_\_\_

8  $10 \times 10$

\_\_\_\_\_

9  $100 \times 10$

\_\_\_\_\_

10 Zeros in ten thousand

\_\_\_\_\_

11 How many tens in 1,000

\_\_\_\_\_

## Core GRADED • SHOW YOUR WORKING

**STOP RULE** Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

12. Write six million forty thousand five (6,040,005 — hold the empty places with zeros)

.....

13. The value of the 5 in 5,400

.....

14. The value of the 8 in 380

.....

★

Challenge

NOT GRADED · REASON IT OUT

15. How many thousands in 10,000?  
.....
16. In 3,472,861 — what is the 4 worth? (It sits in the hundred-thousands place)  
.....
17. In 3,472,861 — what is the 7 worth? (Ten-thousands place)  
.....
18. In 5,208,043 — what is the 2 worth? (Count the places from the right)  
.....
19. How many thousands make 4,000,000? ( $4,000 \times 1,000 = 4,000,000$ )  
.....
20. How many zeros in one million? (1,000,000)  
.....
21.  $1,000 \times 1,000$  (A thousand thousands)  
.....
22. The value of the 2 in 24,000  
.....
23. Thousands in 250,000  
.....
24. Hundreds in 45,000  
.....
25. Millions in one billion  
.....
26. Zeros in a hundred million  
.....



Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE

!

Error journal

ONE SENTENCE PER MISTAKE

# Ten Times Bigger

Multiplying by ten shifts every digit one place left. Watch the zeros.

## Warm-Up 9 ITEMS • RECALL

1  $40 \times 10$   
\_\_\_\_\_

2  $700 \times 10$   
\_\_\_\_\_

3  $60 \times 100$   
\_\_\_\_\_

4  $8 \times 1,000$   
\_\_\_\_\_

5  $3,200 \div 10$   
\_\_\_\_\_

6  $20 \times 10$   
\_\_\_\_\_

7  $500 \times 10$   
\_\_\_\_\_

8  $90 \times 100$   
\_\_\_\_\_

9  $4,000 \div 10$   
\_\_\_\_\_

## Core GRADED • SHOW YOUR WORKING

**STOP RULE** Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

10.  $230 \times 100$  (Two more zeros) .....  
\_\_\_\_\_

11.  $9,000 \div 100$  (Take two zeros away) .....  
\_\_\_\_\_

12.  $350 \times 100$  .....  
\_\_\_\_\_

13.  $1,200 \times 10$  .....  
\_\_\_\_\_

14.  $5,600 \div 10$  .....  
\_\_\_\_\_

★

Challenge

NOT GRADED · REASON IT OUT

15.  $45,000 \div 1,000$
- .....
16.  $56 \times 1,000$  (Three more zeros)
- .....
17.  $740,000 \div 10,000$  (Four zeros off)
- .....
18. How many times bigger is 800,000 than 800? (Three places left)
- .....
19. The 6 in 6,000 is worth how many times the 6 in 60? (Two places apart)
- .....
20. How many 10,000s make one million? ( $1,000,000 \div 10,000$ )
- .....
21.  $7 \times 100$
- .....
22.  $60,000 \div 1,000$
- .....
23.  $84,000 \div 100$
- .....
24.  $25 \times 1,000$
- .....
25. The 3 in 30,000 is worth how many times the 3 in 300
- .....
26. How many 1,000s make ten million
- .....



Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE

!

Error journal

ONE SENTENCE PER MISTAKE

# Compare, Order, Round

Rounding depends on which place you were asked for. Read the question twice.

## Warm-Up 12 ITEMS • RECALL

1 Round 47 to the nearest ten

\_\_\_\_\_

2 Round 83 to the nearest ten

\_\_\_\_\_

3 Round 65 to the nearest ten

\_\_\_\_\_

4 Round 128 to the nearest ten

\_\_\_\_\_

5 Round 254 to the nearest ten

\_\_\_\_\_

6 Round 999 to the nearest ten

\_\_\_\_\_

7 Round 34 to the nearest ten

\_\_\_\_\_

8 Round 76 to the nearest ten

\_\_\_\_\_

9 Round 55 to the nearest ten

\_\_\_\_\_

10 Round 245 to the nearest hundred

\_\_\_\_\_

11 Round 380 to the nearest hundred

\_\_\_\_\_

12 Round 909 to the nearest ten

\_\_\_\_\_

## Core GRADED • SHOW YOUR WORKING

**STOP RULE** Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. Round 2,499 to the nearest hundred (Look at the tens digit only)

\_\_\_\_\_

14. Round 3,472 to the nearest hundred

\_\_\_\_\_

15. Round 6,509 to the nearest thousand

\_\_\_\_\_

★

Challenge

NOT GRADED · REASON IT OUT

16. Round 47,382 to the nearest thousand (382 down beats 618 up)
- .....
17. Round 47,382 to the nearest ten thousand (Now the ends are 40,000 and 50,000)
- .....
18. Round 863,209 to the nearest hundred thousand (63,209 up to 900,000 is the shorter trip)
- .....
19. Larger number: 408,916 or 480,169? Type it. (Compare the ten-thousands place)
- .....
20. Largest number that rounds to 5,000 at the nearest thousand (One less than the halfway point up)
- .....
21. Smallest number that rounds to 5,000 at the nearest thousand (Exactly halfway rounds up)
- .....
22. Round 24,650 to the nearest thousand
- .....
23. Round 89,999 to the nearest ten thousand
- .....
24. Round 12,345 to the nearest hundred
- .....
25. Largest whole number that rounds to 700 at the nearest hundred
- .....
26. Smallest whole number that rounds to 700 at the nearest hundred
- .....



Working space

SHOW THE ROOMS, THE GRID, THE NUMBER LINE

!

Error journal

ONE SENTENCE PER MISTAKE

# Order of Operations

Build the groups first, then add and subtract what's left loose.

## Warm-Up 12 ITEMS • RECALL

1  $3 + 4 \times 2$   
\_\_\_\_\_

2  $10 - 2 \times 3$   
\_\_\_\_\_

3  $12 \div 4 + 5$   
\_\_\_\_\_

4  $6 + 6 \div 2$   
\_\_\_\_\_

5  $2 \times 3 + 4 \times 5$   
\_\_\_\_\_

6  $20 - (4 + 6)$   
\_\_\_\_\_

7  $5 + 2 \times 3$   
\_\_\_\_\_

8  $8 - 3 \times 2$   
\_\_\_\_\_

9  $16 \div 4 + 1$   
\_\_\_\_\_

10  $9 + 9 \div 3$   
\_\_\_\_\_

11  $(5 + 5) \times 2$   
\_\_\_\_\_

12  $30 - (5 + 5)$   
\_\_\_\_\_

## Core GRADED • SHOW YOUR WORKING

**STOP RULE** Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13.  $5 + 3 \times 4$  (Multiply first:  $3 \times 4 = 12$ )  
.....

14.  $(5 + 3) \times 4$  (Brackets overrule everything)  
.....

15.  $18 \div (3 + 6)$  ( $3 + 6 = 9$  first)  
.....

16.  $40 - 6 \times 5 + 2$  ( $40 - 30 + 2$ )  
.....

17.  $3 \times (12 - 4) \div 6$  ( $3 \times 8 = 24$ , then  $\div 6$ )  
.....

18.  $4 \times 5 - 6 \div 2$   
.....

19.  $(12 + 8) \div 4$   
.....

20.  $7 \times (3 + 2)$   
.....

21.  $48 \div (2 \times 3)$   
.....



## SHOW THE ROOMS, THE GRID, THE NUMBER LINE



ONE SENTENCE PER MISTAKE

# Powers & Expression Duel

Enrichment day. The little raised number counts copies, not multiples.

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## Challenge

1.  $2^3 (2 \times 2 \times 2)$

2.  $5^2 (5 \times 5)$

3.  $10^4$  (A 1 with four zeros)

4.  $3^2 + 4^2$  (9 + 16)

5.  $2^5 - 2^3$  (32 - 8)

6. Bigger value:  $4^3$  or  $3^4$ ? Type it. (64 against 81)

7.  $10^6 \div 10^3$  (Six zeros take away three)

8. How many zeros in  $10^8$ ? (The exponent counts them)

9.  $3^3$

10.  $4^2$

11.  $5^3$

12.  $2^5$



## Working space



## Error journal

# Week 1 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

## 1.1 • How Big Is a Million

OUT OF 14

Warm-Up • 1-11 • 10 | 10 | 1000 | 10000 | 10000 | 5 | 7 | 100 | 1000 | 4 | 100

Core • 12-14 • 6040005 | 5000 | 80

Challenge • 15-26 • 10 | 400000 | 70000 | 200000 | 4000 | 6 | 1000000 | 20000 | 250 | 450 | 1000 | 8

## 1.2 • Ten Times Bigger

OUT OF 14

Warm-Up • 1-9 • 400 | 7000 | 6000 | 8000 | 320 | 200 | 5000 | 9000 | 400

Core • 10-14 • 23000 | 90 | 35000 | 12000 | 560

Challenge • 15-26 • 45 | 56000 | 74 | 1000 | 100 | 100 | 700 | 60 | 840 | 25000 | 100 | 10000

## 1.3 • Compare, Order, Round

OUT OF 15

Warm-Up • 1-12 • 50 | 80 | 70 | 130 | 250 | 1000 | 30 | 80 | 60 | 200 | 400 | 910

Core • 13-15 • 2500 | 3500 | 7000

Challenge • 16-26 • 47000 | 50000 | 900000 | 480169 | 5499 | 4500 | 25000 | 90000 | 12300 | 749 | 650

## 1.4 • Order of Operations

OUT OF 21

Warm-Up • 1-12 • 11 | 4 | 8 | 9 | 26 | 10 | 11 | 2 | 5 | 12 | 20 | 20

Core • 13-21 • 17 | 32 | 2 | 12 | 4 | 17 | 5 | 35 | 8

Challenge • 22-26 • 11 | 36 | 40 | 50 | 14

## Fri • Powers & Expression Duel

ENRICHMENT

Challenge • 1-14 • 8 | 25 | 10000 | 25 | 24 | 81 | 1000 | 8 | 27 | 16 | 125 | 32 | 36 | 64

# Rounding to Any Place

Find the place, look one to its right, decide.

## Warm-Up 12 ITEMS • RECALL

1 Round 48 to the nearest ten

\_\_\_\_\_

2 Round 43 to the nearest ten

\_\_\_\_\_

3 Round 250 to the nearest hundred

\_\_\_\_\_

4 Round 149 to the nearest hundred

\_\_\_\_\_

5 Round 75 to the nearest ten

\_\_\_\_\_

6 Round 612 to the nearest ten

\_\_\_\_\_

7 Round 62 to the nearest ten

\_\_\_\_\_

8 Round 67 to the nearest ten

\_\_\_\_\_

9 Round 340 to the nearest hundred

\_\_\_\_\_

10 Round 360 to the nearest hundred

\_\_\_\_\_

11 Round 95 to the nearest ten

\_\_\_\_\_

12 Round 450 to the nearest hundred

\_\_\_\_\_

## Core GRADED • SHOW YOUR WORKING

**STOP RULE** Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. Round 4,829 to the nearest hundred

\_\_\_\_\_

14. Round 4,829 to the nearest thousand

\_\_\_\_\_

15. Round 8,950 to the nearest hundred

\_\_\_\_\_

16. Round 5,432 to the nearest thousand

\_\_\_\_\_

17. Round 5,632 to the nearest thousand

\_\_\_\_\_

18. Round 7,777 to the nearest hundred

\_\_\_\_\_



SHOW THE ROOMS, THE GRID, THE NUMBER LINE



ONE SENTENCE PER MISTAKE

# Estimate Before You Compute

Round first. Then the real answer has something to be checked against.

## Warm-Up 11 ITEMS • RECALL

1 Estimate  $48 + 31$  by rounding both to tens

\_\_\_\_\_

2 Estimate  $62 - 19$

\_\_\_\_\_

3 Estimate  $19 \times 21$

\_\_\_\_\_

4 Estimate  $81 \div 9$

\_\_\_\_\_

5 Estimate  $297 + 104$

\_\_\_\_\_

6 Estimate  $52 + 48$

\_\_\_\_\_

7 Estimate  $39 + 42$  by rounding to tens

\_\_\_\_\_

8 Estimate  $71 - 28$

\_\_\_\_\_

9 Estimate  $29 \times 11$

\_\_\_\_\_

10 Estimate  $63 \div 8$

\_\_\_\_\_

11 Estimate  $198 + 205$

\_\_\_\_\_

## Core GRADED • SHOW YOUR WORKING

**STOP RULE** Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

12. Estimate  $4,812 + 3,190$  to the nearest thousand

.....

13. Estimate  $612 - 388$  to the nearest hundred

.....

14. Estimate  $3,964 \div 4$

.....

15. Estimate  $486 + 317$  to the nearest hundred

.....

16. Estimate  $4,120 - 1,890$  to the nearest hundred

.....

17. Estimate  $396 \times 21$

.....

18. Estimate  $5,940 \div 6$

.....

19. Estimate  $78 \times 41$

.....

**Challenge** NOT GRADED • REASON IT OUT

20. Estimate  $187 \times 4$

21. True value of  $187 \times 4$

22. Estimate the cost of 19 items at \$4.95 each

23. Someone estimates  $512 \times 6$  as 300. Type the real answer

24. Estimate  $52 \times 9$

25. Estimate 12 items at \$19.95

26. Someone estimates  $407 \times 8$  as 320. Type the real answer

 **Working space** SHOW THE ROOMS, THE GRID, THE NUMBER LINE

**Error journal** ONE SENTENCE PER MISTAKE

# How Wrong Is the Estimate

An estimate you never compare to anything teaches you nothing.

## Warm-Up 10 ITEMS • RECALL

1 Estimate of  $48 + 31$ 

\_\_\_\_\_

2 True value of  $48 + 31$ 

\_\_\_\_\_

3 The difference

\_\_\_\_\_

4 True value of  $19 \times 21$ 

\_\_\_\_\_

5 Estimate of  $19 \times 21$ 

\_\_\_\_\_

6 Estimate of  $39 + 42$ 

\_\_\_\_\_

7 True value of  $39 + 42$ 

\_\_\_\_\_

8 True value of  $29 \times 11$ 

\_\_\_\_\_

9 Estimate of  $29 \times 11$ 

\_\_\_\_\_

10 The gap between them

\_\_\_\_\_

## Core GRADED • SHOW YOUR WORKING

**STOP RULE** Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

11. True value of  $612 - 388$ 

.....

12. Estimate of  $612 - 388$  to the nearest hundred

.....

13. How far off was that estimate

.....

14. True value of  $4,812 + 3,190$ 

.....

15. How far off was the thousand estimate

.....

16. Estimate of  $486 + 317$ 

.....

17. True value of  $486 + 317$ 

.....

18. The difference there

.....

19. True value of  $78 \times 41$ 

.....

20. Estimate of  $78 \times 41$ 

.....



# Big Numbers in Context

Millions stop being abstract when you attach them to something real.

## Warm-Up 11 ITEMS • RECALL

1 Zeros in one thousand

\_\_\_\_\_

2 Zeros in one million

\_\_\_\_\_

3 Thousands in a million

\_\_\_\_\_

4 Hundreds in a thousand

\_\_\_\_\_

5 Tens in a thousand

\_\_\_\_\_

6 Zeros in ten thousand

\_\_\_\_\_

7 Zeros in a hundred thousand

\_\_\_\_\_

8 Ten thousands in a million

\_\_\_\_\_

9 Hundreds in ten thousand

\_\_\_\_\_

10 Tens in ten thousand

\_\_\_\_\_

11 Thousands in ten thousand

\_\_\_\_\_

## Core GRADED • SHOW YOUR WORKING

**STOP RULE** Do the first **five**. All five right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

12. Days in about a million minutes, rounded to the nearest hundred (A million minutes is about 694 days)

.....

13. Dots per page 1,000 — pages for a million dots

.....

14. Seconds in an hour

.....

15. Hours in a week

.....

16. Minutes in a day

.....

17. Days in ten years, ignoring leap days

.....

18. Seconds in a day

.....



# A Million Dots, Measured

Prove the estimate. Count one square, scale it up, defend the number.

## ✦ Warm-Up 4 ITEMS • RECALL

1 Dots in a 10 by 10 square

\_\_\_\_\_

2 Dots in ten such squares

\_\_\_\_\_

3 Dots in a 5 by 5 square

\_\_\_\_\_

4 Dots in four such squares

\_\_\_\_\_

## ◆ Core GRADED • SHOW YOUR WORKING

**STOP RULE** Short Core today — do all of them, then turn the page over.

5. Squares of 100 dots needed for a million

.....

6. A 100 by 100 grid holds how many dots

.....

7. Dots in a 20 by 20 square

.....



# Week 2 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

## 2.1 • Rounding to Any Place

OUT OF 18

Warm-Up • 1-12 • 50 | 40 | 300 | 100 | 80 | 610 | 60 | 70 | 300 | 400 | 100 | 500

Core • 13-18 • 4800 | 5000 | 9000 | 5000 | 6000 | 7800

Challenge • 19-26 • 28000 | 100000 | 250 | 349 | 19000 | 46000 | 7500 | 8499

## 2.2 • Estimate Before You Compute

OUT OF 19

Warm-Up • 1-11 • 80 | 40 | 400 | 9 | 400 | 100 | 80 | 40 | 300 | 8 | 400

Core • 12-19 • 8000 | 200 | 1000 | 800 | 2200 | 8000 | 1000 | 3200

Challenge • 20-26 • 800 | 748 | 100 | 3072 | 500 | 240 | 3256

## 2.3 • How Wrong Is the Estimate

OUT OF 20

Warm-Up • 1-10 • 80 | 79 | 1 | 399 | 400 | 80 | 81 | 319 | 300 | 19

Core • 11-20 • 224 | 200 | 24 | 8002 | 2 | 800 | 803 | 3 | 3198 | 3200

Challenge • 21-24 • high | 1599 | low | 2499

## 2.4 • Big Numbers in Context

OUT OF 18

Warm-Up • 1-11 • 3 | 6 | 1000 | 10 | 100 | 4 | 5 | 100 | 100 | 1000 | 10

Core • 12-18 • 700 | 1000 | 3600 | 168 | 1440 | 3650 | 86400

Challenge • 19-22 • 3000 | 100 | 99 | 12

## Fri • A Million Dots, Measured

OUT OF 8

Warm-Up • 1-4 • 100 | 1000 | 25 | 100

Core • 5-8 • 10000 | 10000 | 400 | 100

Challenge • 9-14 • 100 | 20 | 1000000 | 500 | 50000 | 100

# Why Order Matters

Without an agreed order, one expression has several answers.

## Warm-Up 12 ITEMS • RECALL

1  $2 + 3 \times 4$   
\_\_\_\_\_

2  $(2 + 3) \times 4$   
\_\_\_\_\_

3  $10 - 2 \times 3$   
\_\_\_\_\_

4  $(10 - 2) \times 3$   
\_\_\_\_\_

5  $6 + 6 \div 2$   
\_\_\_\_\_

6  $(6 + 6) \div 2$   
\_\_\_\_\_

7  $3 + 2 \times 5$   
\_\_\_\_\_

8  $(3 + 2) \times 5$   
\_\_\_\_\_

9  $12 - 3 \times 2$   
\_\_\_\_\_

10  $(12 - 3) \times 2$   
\_\_\_\_\_

11  $8 + 8 \div 4$   
\_\_\_\_\_

12  $(8 + 8) \div 4$   
\_\_\_\_\_

## Core GRADED • SHOW YOUR WORKING

**STOP RULE** Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13.  $5 + 4 \times 3 - 2$  .....  
\_\_\_\_\_

14.  $(5 + 4) \times (3 - 2)$  .....  
\_\_\_\_\_

15.  $20 - 3 \times 4 + 2$  .....  
\_\_\_\_\_

16.  $36 \div (2 + 4)$  .....  
\_\_\_\_\_

17.  $36 \div 2 + 4$  .....  
\_\_\_\_\_

18.  $20 - 3 \times 4 + 1$  .....  
\_\_\_\_\_

19.  $(20 - 3) \times 4$  .....  
\_\_\_\_\_

20.  $2 \times 6 + 3 \times 4$  .....  
\_\_\_\_\_

21.  $2 \times (6 + 3) \times 4$  .....  
\_\_\_\_\_

22.  $36 \div 6 \div 2$  .....  
\_\_\_\_\_



# Brackets First

Whatever is inside the brackets happens before anything else.

## Warm-Up 12 ITEMS • RECALL

1  $(4 + 5) \times 2$   
\_\_\_\_\_

2  $(9 - 3) \div 2$   
\_\_\_\_\_

3  $3 \times (2 + 2)$   
\_\_\_\_\_

4  $(7 + 3) \times 10$   
\_\_\_\_\_

5  $(12 - 4) \div 4$   
\_\_\_\_\_

6  $5 \times (10 - 6)$   
\_\_\_\_\_

7  $(2 + 6) \times 3$   
\_\_\_\_\_

8  $(15 - 5) \div 5$   
\_\_\_\_\_

9  $4 \times (3 + 3)$   
\_\_\_\_\_

10  $(9 + 1) \times 10$   
\_\_\_\_\_

11  $(20 - 8) \div 4$   
\_\_\_\_\_

12  $6 \times (12 - 8)$   
\_\_\_\_\_

## Core GRADED • SHOW YOUR WORKING

**STOP RULE** Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13.  $(15 + 5) \div (2 + 2)$  .....

14.  $4 \times (6 + 3) - 10$  .....

15.  $(8 \times 3) - (4 \times 2)$  .....

16.  $2 \times (50 - 15)$  .....

17.  $(25 - 9) \div (2 \times 4)$  .....

18.  $((3 + 3) \times 4) - 10$  .....

19.  $100 \div (5 \times 5)$  .....

20.  $(7 + 8) \times (10 - 6)$  .....

21.  $(45 \div 9) + (6 \times 3)$  .....

**Challenge** NOT GRADED • REASON IT OUT

22.  $100 - (25 \times 3)$

23.  $((4 + 2) \times 3) - 8$

24.  $120 \div (2 \times (3 + 3))$

25.  $((10 + 2) \times 5) \div 3$

26.  $240 \div (4 \times (2 + 1))$

 **Working space** SHOW THE ROOMS, THE GRID, THE NUMBER LINE

## ! Error journal ONE SENTENCE PER MISTAKE

# Multi-Step Expressions

Multiply and divide before you add and subtract. Left to right within each.

## Warm-Up 12 ITEMS • RECALL

1  $12 \div 3 + 2$   
\_\_\_\_\_

2  $12 + 3 \times 2$   
\_\_\_\_\_

3  $20 \div 4 \times 2$   
\_\_\_\_\_

4  $18 - 6 \div 3$   
\_\_\_\_\_

5  $7 \times 2 + 6$   
\_\_\_\_\_

6  $30 \div 5 - 2$   
\_\_\_\_\_

7  $18 \div 3 + 4$   
\_\_\_\_\_

8  $18 + 3 \times 4$   
\_\_\_\_\_

9  $24 \div 6 \times 2$   
\_\_\_\_\_

10  $20 - 8 \div 4$   
\_\_\_\_\_

11  $9 \times 3 + 3$   
\_\_\_\_\_

12  $40 \div 8 - 3$   
\_\_\_\_\_

## Core GRADED • SHOW YOUR WORKING

**STOP RULE** Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

13.  $48 \div 6 + 3 \times 4$   
.....

14.  $7 \times 8 - 6 \times 5$   
.....

15.  $64 \div 8 \div 2$   
.....

16.  $9 + 36 \div 4 - 3$   
.....

17.  $60 \div 5 + 7 \times 2$   
.....

18.  $81 \div 9 - 3 \times 2$   
.....

19.  $7 \times 8 + 100 \div 4$   
.....



## Challenge

20.  $100 - 4 \times 12$

21.  $144 \div 12 + 8 \times 3 - 5$

22.  $2 \times 3 + 4 \times 5 + 6 \times 7$

23.  $11 \times 4 - 22 \div 2$

24.  $144 \div 12 \times 3$

25.  $200 \div 8 + 6 \times 7 - 9$

26.  $3 \times 4 + 5 \times 6 + 7 \times 8$



## Working space



## Error journal

# Write the Expression

Turn the sentence into symbols, then work it out.

## Warm-Up 12 ITEMS • RECALL

1 Five more than three lots of four

\_\_\_\_\_

2 Double seven, then add one

\_\_\_\_\_

3 Ten less than six times three

\_\_\_\_\_

4 Half of twenty, plus four

\_\_\_\_\_

5 Three lots of nine

\_\_\_\_\_

6 Twenty shared by four

\_\_\_\_\_

7 Four more than double six

\_\_\_\_\_

8 Triple five, then add two

\_\_\_\_\_

9 Twenty less than five times six

\_\_\_\_\_

10 A third of thirty, plus one

\_\_\_\_\_

11 Two lots of nine, minus eight

\_\_\_\_\_

12 Half of sixteen, times three

\_\_\_\_\_

## Core GRADED • SHOW YOUR WORKING

**STOP RULE** Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

13. The sum of 8 and 4, multiplied by 3

.....

14. 8 added to 4 lots of 3

.....

15. 100 less the product of 7 and 9

.....

16. The difference of 20 and 8, divided by 4

.....

17. Six lots of the sum of 2 and 3

.....

18. Six times the sum of two and five

.....

19. The product of 7 and 8, less 20

.....

20. Divide 96 by the sum of 4 and 8

.....

21. Add 15 to the product of 6 and 9

.....

22. Double the difference between 30 and 12

.....



# Mid-Unit Quiz

Eight items across Weeks 1–3. 85% to keep flying.

## ✦ Warm-Up 4 ITEMS • RECALL

1 Round 4,829 to the nearest hundred  
\_\_\_\_\_2  $2 + 3 \times 4$   
\_\_\_\_\_3 Round 3,451 to the nearest hundred  
\_\_\_\_\_4  $5 + 2 \times 3$   
\_\_\_\_\_

## ◆ Core GRADED • SHOW YOUR WORKING

**STOP RULE** This one is a test. Do **every** problem — no stopping early.5.  $(5 + 4) \times (3 - 2)$  .....6.  $48 \div 6 + 3 \times 4$  .....

7. Zeros in one million .....

8. The sum of 8 and 4, multiplied by 3 .....

9.  $(20 - 3) \times 4$  .....10. Estimate  $396 \times 21$  .....



# Week 3 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

## 3.1 • Why Order Matters

OUT OF 22

Warm-Up • 1-12 • 14 | 20 | 4 | 24 | 9 | 6 | 13 | 25 | 6 | 18 | 10 | 4

Core • 13-22 • 15 | 9 | 10 | 6 | 22 | 9 | 68 | 24 | 72 | 3

Challenge • 23-26 • 70 | 50 | 54 | 30

## 3.2 • Brackets First

OUT OF 21

Warm-Up • 1-12 • 18 | 3 | 12 | 100 | 2 | 20 | 24 | 2 | 24 | 100 | 3 | 24

Core • 13-21 • 5 | 26 | 16 | 70 | 2 | 14 | 4 | 60 | 23

Challenge • 22-26 • 25 | 10 | 10 | 20 | 20

## 3.3 • Multi-Step Expressions

OUT OF 19

Warm-Up • 1-12 • 6 | 18 | 10 | 16 | 20 | 4 | 10 | 30 | 8 | 18 | 30 | 2

Core • 13-19 • 20 | 26 | 4 | 15 | 26 | 3 | 81

Challenge • 20-26 • 52 | 31 | 68 | 33 | 36 | 58 | 98

## 3.4 • Write the Expression

OUT OF 22

Warm-Up • 1-12 • 17 | 15 | 8 | 14 | 27 | 5 | 16 | 17 | 10 | 11 | 10 | 24

Core • 13-22 • 36 | 20 | 37 | 3 | 30 | 42 | 36 | 8 | 69 | 36

Challenge • 23-26 • 24 | 4 | 48 | 10

## Fri • Mid-Unit Quiz

OUT OF 11

Warm-Up • 1-4 • 4800 | 14 | 3500 | 11

Core • 5-11 • 9 | 20 | 800 | 6 | 36 | 68 | 8000

Challenge • 12-15 • 31 | 14 | 749 | 86400

# The Little Raised Number

An exponent counts how many times the base is multiplied by itself.

## ✦ Warm-Up 10 ITEMS • RECALL

1

 $2^2$ 

\_\_\_\_\_

2

 $3^2$ 

\_\_\_\_\_

3

 $5^2$ 

\_\_\_\_\_

4

 $2^3$ 

\_\_\_\_\_

5

 $4^2$ 

\_\_\_\_\_

6

 $6^2$ 

\_\_\_\_\_

7

 $7^2$ 

\_\_\_\_\_

8

 $8^2$ 

\_\_\_\_\_

9

 $9^2$ 

\_\_\_\_\_

10

 $1^5$ 

\_\_\_\_\_

## ◆ Core GRADED • SHOW YOUR WORKING

**STOP RULE** Do the first **five**. All five right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

11.  $3^3$ 

.....

12.  $2^4$ 

.....

13.  $5^3$ 

.....

14.  $2^5$ 

.....

15.  $4^3$ 

.....

16.  $11^2$ 

.....

**Challenge** NOT GRADED • REASON IT OUT

17.  $10^2$

18.  $10^3$

19.  $2^8$

20. Which is bigger,  $3^4$  or  $4^3$  — type the value of the bigger

21.  $2^{10}$

22. Which is bigger,  $2^6$  or  $6^2$  — type the value of the bigger

 **Working space** SHOW THE ROOMS, THE GRID, THE NUMBER LINE

**Error journal** ONE SENTENCE PER MISTAKE

# Powers of Ten

The exponent is the number of zeros. That is the whole shortcut.

## Warm-Up 3 ITEMS • RECALL

1  $10^1$ 2 Zeros in  $10^7$ 3  $10^0$ 

## Core GRADED • SHOW YOUR WORKING

**STOP RULE** Do the first **five**. All five right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

4. Write 5,000 as a digit times a power of ten — type the exponent

.....

**Challenge** NOT GRADED • REASON IT OUT

5.  $10^2$

6.  $10^3$

7.  $10^4$

8. Zeros in  $10^5$

9.  $10^6$

10.  $3 \times 10^2$

11.  $7 \times 10^3$

12.  $4 \times 10^4$

13.  $25 \times 10^2$

14.  $10^3 \times 10^2$

15. A million written as a power of ten — type the exponent

16.  $10^5$

 **Working space** SHOW THE ROOMS, THE GRID, THE NUMBER LINE

**Error journal** ONE SENTENCE PER MISTAKE

# Expanded Form with Powers

Every place value is a power of ten in disguise.

## Warm-Up 12 ITEMS • RECALL

1 The value of the 4 in 400

\_\_\_\_\_

2 The value of the 7 in 70

\_\_\_\_\_

3 The value of the 3 in 3,000

\_\_\_\_\_

4  $200 + 30 + 5$

\_\_\_\_\_

5  $4,000 + 200 + 10$

\_\_\_\_\_

6 The value of the 9 in 900

\_\_\_\_\_

7 How many hundreds make 2,600

\_\_\_\_\_

8 How many hundreds make 4,300

\_\_\_\_\_

9 How many hundreds make 7,100

\_\_\_\_\_

10 How many hundreds make 9,500

\_\_\_\_\_

11 How many tens make 3,000

\_\_\_\_\_

12 How many tens make 6,000

\_\_\_\_\_

## Core GRADED • SHOW YOUR WORKING

**STOP RULE** Do the first **five**. All five right? Stop and turn the page over. Missed any? Keep going until you get **five** right in a row.

20. In 4,271 — what is the 4 worth

.....

21. In 4,271 — what is the 2 worth

.....

22. In 4,271 — what is the 7 worth

.....

23. In 8,605 — what is the 8 worth

.....

24. In 8,605 — what is the 6 worth

.....

25. In 3,940 — what is the 9 worth

.....

26. In 7,128 — what is the 7 worth

.....

27. In 7,128 — what is the 1 worth

.....

28. In 5,083 — what is the 5 worth

.....

29. Bigger: 3,421 or 3,412

.....

30. Bigger: 5,680 or 5,806

.....



# Error Journal Sweep

Mixed review of the whole mission. Fix only what repeats.

## Warm-Up 10 ITEMS • RECALL

1 Round 149 to the nearest hundred  
\_\_\_\_\_

2  $2 + 3 \times 4$   
\_\_\_\_\_

3 Zeros in one million  
\_\_\_\_\_

4  $5^2$   
\_\_\_\_\_

5  $(2 + 3) \times 4$   
\_\_\_\_\_

6 Round 340 to the nearest hundred  
\_\_\_\_\_

7  $5 + 2 \times 3$   
\_\_\_\_\_

8 Zeros in ten thousand  
\_\_\_\_\_

9  $6^2$   
\_\_\_\_\_

10  $(3 + 2) \times 5$   
\_\_\_\_\_

## Core GRADED • SHOW YOUR WORKING

**STOP RULE** Do the first **six**. All six right? Stop and turn the page over. Missed any? Keep going until you get **five right in a row**.

11.  $48 \div 6 + 3 \times 4$  .....

12. Estimate  $3,964 \div 4$  .....

13. Estimate  $486 + 317$  to the nearest hundred .....

14.  $60 \div 5 + 7 \times 2$  .....

15.  $2^4$  .....



26.  $2^{10}$



# Mission 03 Test

Twelve items plus the Big Question, answered out loud.

## Warm-Up 4 ITEMS • RECALL

1 Round 612 to the nearest ten  
\_\_\_\_\_2  $3^2$   
\_\_\_\_\_3 Round 448 to the nearest ten  
\_\_\_\_\_4  $4^2$   
\_\_\_\_\_

## Core GRADED • SHOW YOUR WORKING

**STOP RULE** This one is a test. Do **every** problem — no stopping early.5. Round 4,829 to the nearest thousand  
.....6.  $10 - 2 \times 3$   
.....7.  $(15 + 5) \div (2 + 2)$   
.....8.  $2^5$   
.....9. Estimate  $4,812 + 3,190$  to the nearest thousand  
.....10. The sum of 12 and 8, divided by the difference of 9 and 4  
.....11.  $(15 - 5) \div 5$   
.....



SHOW THE ROOMS, THE GRID, THE NUMBER LINE



ONE SENTENCE PER MISTAKE

# Week 4 Answers

Numbers run straight through each sheet, front to back. Score the Warm-Up and Core the child actually attempted — Challenge never counts.

Mark the reasoning, not the handwriting.

## 4.1 • The Little Raised Number

OUT OF 16

Warm-Up • 1-10 • 4 | 9 | 25 | 8 | 16 | 36 | 49 | 64 | 81 | 1

Core • 11-16 • 27 | 16 | 125 | 32 | 64 | 121

Challenge • 17-22 • 100 | 1000 | 256 | 81 | 1024 | 64

## 4.2 • Powers of Ten

OUT OF 4

Warm-Up • 1-3 • 10 | 7 | 1

Core • 4 • 3

Challenge • 5-16 • 100 | 1000 | 10000 | 5 | 1000000 | 300 | 7000 | 40000 | 2500 |  
100000 | 6 | 100000

## 4.3 • Expanded Form with Powers

OUT OF 23

Warm-Up • 1-12 • 400 | 70 | 3000 | 235 | 4210 | 900 | 26 | 43 | 71 | 95 | 300 |  
600

Core • 20-30 • 4000 | 200 | 70 | 8000 | 600 | 900 | 7000 | 100 | 5000 | 3,421 |  
5,806

Challenge • 33-44 • 6402 | 30500 | 80000 | 999 | 700000 | 206040 | 4 | 5200 | 70300 |  
4062 | 24350 | 307050

## Thu • Error Journal Sweep

OUT OF 22

Warm-Up • 1-12 • 100 | 14 | 1000 | 6 | 25 | 20 | 300 | 11 | 10000 | 4 | 36 | 25

Core • 13-22 • 100000 | 20 | 40000 | 1000 | 6402 | 800 | 26 | 25000 | 4062 | 16

Challenge • 23-26 • 31 | 256 | 58 | 1024

## Fri • Mission 03 Test

OUT OF 14

Warm-Up • 1-4 • 610 | 9 | 450 | 16

Core • 5-14 • 5000 | 4 | 5 | 52 | 7000 | 32 | 8000 | 4 | 2 | 70000

Challenge • 15-19 • 206040 | 68 | 98 | 12 | 125